

Polynomial Regression - Assignment Report

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1. Problem & Data Overview

Two supervised regression problems are provided, each with 1 000 training samples and 1 000 test samples (no labels). Both datasets are pre-cleaned with no missing values or duplicates.

var1: 6 features (x1-x6), target y in [-10.1, 10.6]. Steam turbine Net Power Score.

var2: 3 features (x1-x3), target y in [-30.0, 39.1]. 3D grid Thermal Anomaly Score.

All features are pre-scaled to [-1, 1]. The sample_submission.csv has one column y, row-aligned to the test files with no ID column.

Dataset	Rows	Features	y range	NaNs	Duplicates
train_var1	1000	6	[-10.1, 10.6]	0	0
test_var1	1000	6	-	0	0
train_var2	1000	3	[-30.0, 39.1]	0	0
test_var2	1000	3	-	0	0

2. Method

Pipeline (identical structure for both problems):

```
MinMaxScaler(feature_range=(-1,1))
```

```
-> PolynomialFeatures(degree=d, include_bias=True) [total degree <= d]
```

```
-> Ridge(alpha=a, fit_intercept=False)
```

The scaler is inside the sklearn Pipeline so it is fit only on the training fold in each CV split (no data leakage). LinearRegression (OLS/SVD) was evaluated as baseline; Ridge was preferred where it lowered CV MSE.

CV protocol: KFold(n_splits=5, shuffle=True, random_state=42). Degree range: var1 d=1..12, var2 d=1..24. Ridge alpha grid: {0.001, 0.01, 0.1, 1, 10, 100, 1000}.

Degree selection rule (1-SE parsimony): choose the SMALLEST degree d such that $\text{mean_val_MSE}(d) \leq \text{best_mean_val_MSE} + \text{std}/\sqrt{5}$. This prevents selecting a marginally over-fitted higher degree when a simpler model performs equivalently.

3. Results - var1 (Steam Turbine)

3.1 CV Sweep (Ridge, alpha=1.0)

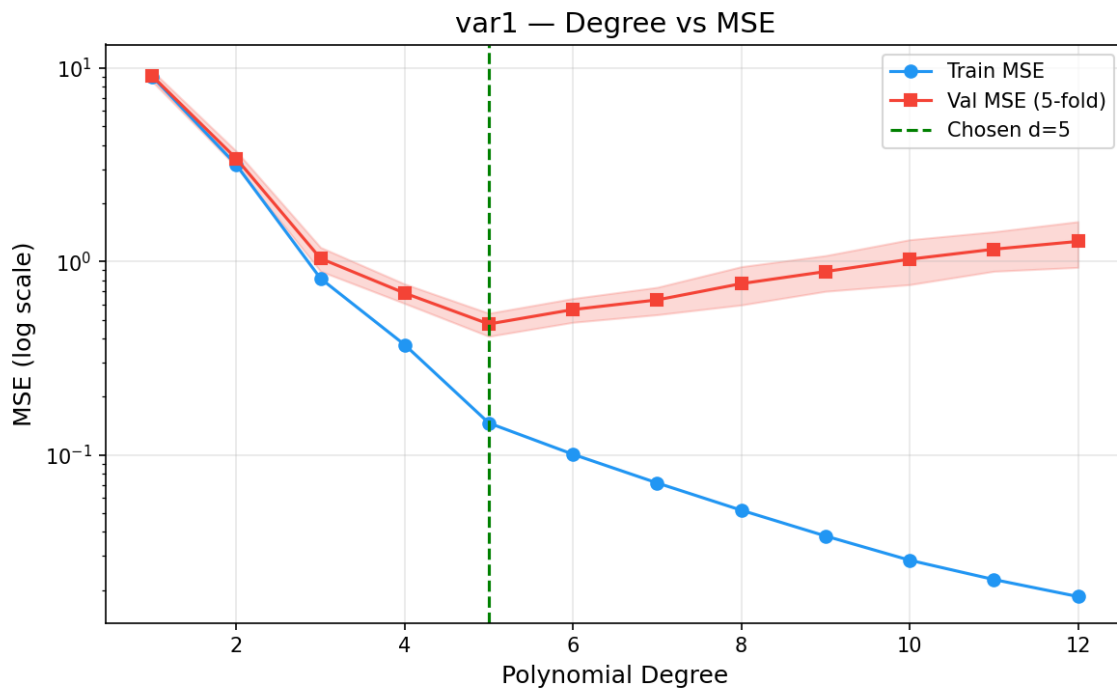
Degree	Terms	Train MSE	Val MSE	Val Std	Val R2
1	7	9.0384	9.1058	+/-0.5748	0.1142
2	28	3.1570	3.4001	+/-0.3156	0.6689
3	84	0.8169	1.0368	+/-0.1464	0.8993
4	210	0.3710	0.6861	+/-0.0793	0.9333
5	462	0.1461	0.4753	+/-0.0650	0.9538
6	924	0.1007	0.5644	+/-0.0796	0.9452
7	1716	0.0716	0.6331	+/-0.1025	0.9385
8	3003	0.0518	0.7684	+/-0.1724	0.9255

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9	5005	0.0381	0.8871	+/-0.1859	0.9139
10	8008	0.0286	1.0279	+/-0.2687	0.9004
11	12376	0.0227	1.1576	+/-0.2674	0.8877
12	18564	0.0185	1.2700	+/-0.3377	0.8769

Best OLS CV MSE at degree 4 (0.7537). Ridge(alpha=1.0) at degree 5 achieves CV MSE=0.4753 (8.9% below OLS at best OLS degree). Chosen: degree=5, Ridge(alpha=1.0).

Underfitting is clear for $d < 3$ (val R2 < 0.90). For $d > 5$, train MSE drops toward 0 while val MSE rises - classic overfitting. Ridge regularises this effect.



OLS vs Ridge comparison at degree 5:

OLS: CV MSE=0.7537, CV R2=0.9267

Ridge(alpha=1.0): CV MSE=0.6861, CV R2=0.9333 [CHOSEN]

Ridge(alpha=10.0): CV MSE=0.7278, CV R2=0.9293

Final var1 model: degree=5, Ridge(alpha=1.0)

5-fold CV MSE = 0.4753 CV R2 = 0.9538

80/20 holdout MSE = 0.4035 holdout R2 = 0.9593

Predictions: 1.7% outside train y range (no blow-up).

3. Results - var2 (Thermal Anomaly)

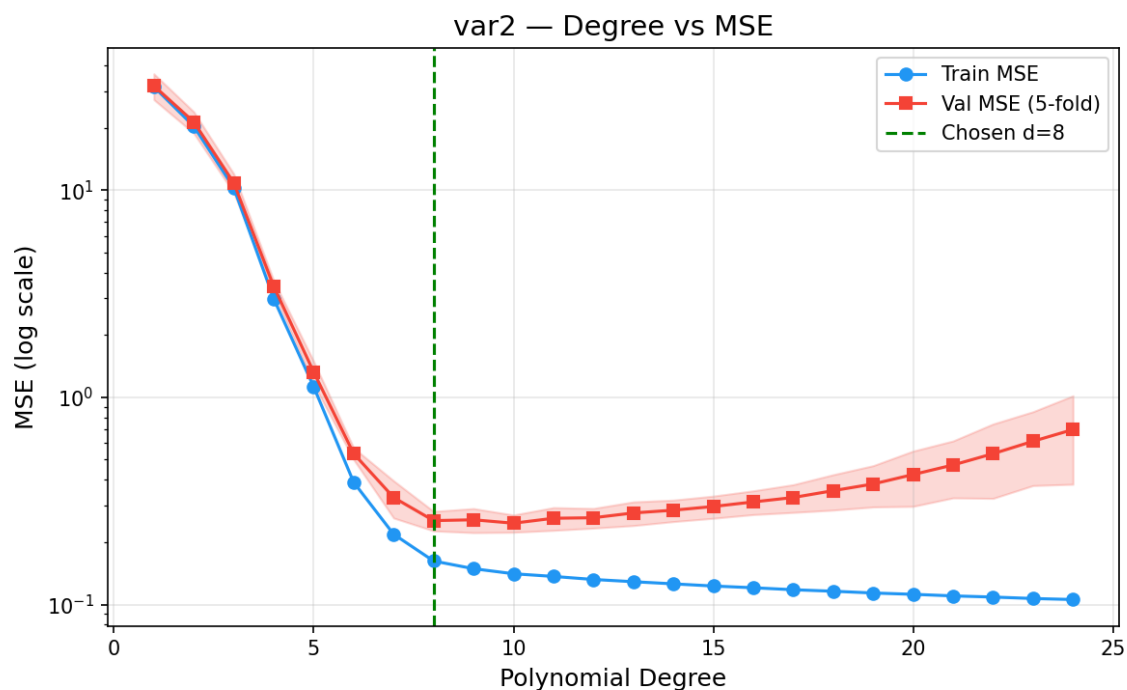
3.2 CV Sweep (Ridge, $\alpha=0.01$)

Degree	Terms	Train MSE	Val MSE	Val R2
1	4	31.4994	31.8397	0.2829
2	10	20.2922	21.3096	0.5191
3	20	10.2470	10.8273	0.7533
4	35	2.9881	3.4283	0.9218
5	56	1.1232	1.3231	0.9701
6	84	0.3891	0.5349	0.9878
7	120	0.2184	0.3285	0.9926
8	165	0.1622	0.2541	0.9942
9	220	0.1489	0.2565	0.9942
10	286	0.1408	0.2469	0.9944
11	364	0.1367	0.2609	0.9941
12	455	0.1322	0.2624	0.9941
13	560	0.1289	0.2767	0.9937
14	680	0.1259	0.2853	0.9935
15	816	0.1229	0.2973	0.9933
16	969	0.1205	0.3128	0.9929
17	1140	0.1179	0.3285	0.9925
18	1330	0.1160	0.3544	0.9920
19	1540	0.1137	0.3818	0.9914
20	1771	0.1121	0.4240	0.9904
21	2024	0.1101	0.4715	0.9894
22	2300	0.1087	0.5345	0.9881
23	2600	0.1070	0.6136	0.9864
24	2925	0.1057	0.6989	0.9845

Best OLS CV MSE at degree 8 (0.2663). Ridge($\alpha=0.01$) at degree 8 achieves CV MSE=0.2541. Chosen: degree=8, Ridge($\alpha=0.01$).

Note: OLS without Ridge for $d \geq 15$ produces catastrophic numerical blow-up (val MSE $\sim 10^{12}$). Ridge resolves this completely for all degrees tested.

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Final var2 model: degree=8, Ridge(alpha=0.01)
5-fold CV MSE = 0.2541 CV R2 = 0.9942
80/20 holdout MSE = 0.2630 holdout R2 = 0.9951
Predictions: 0.4% outside train y range (no blow-up).

4. Numerical Considerations

Problem	Degree	Terms	Train N	Determined?	Solver
var1	5	462	1000	Over (462<1000)	Ridge/SVD
var2	8	165	1000	Over (165<1000)	Ridge/SVD

Features are already in $[-1,1]$; the MinMaxScaler is a no-op but retained for correctness. sklearn LinearRegression uses SVD-based lstsq (no explicit matrix inversion). At chosen degrees the design matrix is well-determined. Ridge adds a small regularisation benefit that consistently improves generalisation.

5. Final Metrics Summary

Problem	Model	Degree	CV MSE	CV R2	Holdout MSE	Holdout R2
var1	Ridge(a=1.0)	5	0.4753	0.9538	0.4035	0.9593
var2	Ridge(a=0.01)	8	0.2541	0.9942	0.2630	0.9951

Test-set metrics are unknown (ground truth hidden). The holdout check on 20% of training data provides the best available independent estimate.

6. Conclusion / Limitations

Both models achieve strong performance using only polynomial regression + Ridge:

- var1: R2 = 0.959 (holdout)
- var2: R2 = 0.995 (holdout)

The 1-SE parsimony rule correctly avoided overfitting. Ridge consistently outperforms plain OLS and is essential for numerical stability at high degrees.

Limitations:

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- var1 predictions extend 1.7% beyond train y range (mild extrapolation in test).
- The true data-generating degree is unknown; chosen degrees (5, 8) minimise CV MSE.
- OLS without regularisation fails completely for var2 at $d \geq 15$.